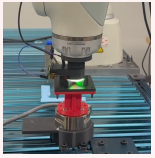
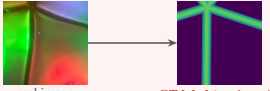
(a) Register real contacts



each label needs its own registered press (q^*)

labels grow only as fast as registration; repeated for every object and sensor

project mesh at q^*

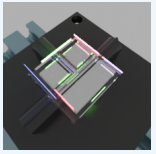


real image

GT label (mesh at q^*)

cost: one registered pose per label

(b) Simulate contacts

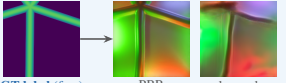


geometry is free, but the appearance must be rendered

LED colors, gel optics and color response fitted per sensor, or a renderer trained on real image–label pairs; a gap remains

render

same contact as (a)



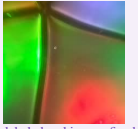
GT label (free)

PBR

learned

cost: calibration or real pairs; appearance gap remains

(c) AlignTac: align the mesh to the real image



unlabeled real image, fixed

learned from the K registered contacts by perturbing their q^*



aligner $\mathcal{A}$ moves the mesh state q

align



mesh at teacher pose

pseudo-label (mesh at aligned q)

cost: K GT labels; the rest get pseudo-labels